Smart Rope Slingshot for Space Launch V2.0

Feasibility Study: Electrically Actuated Ropes for High-Altitude Balloon Launch Systems

Version 2.0 — Revised & Expanded | June 2026

Δ Change Log

- Harvard Wyss Group 2025 soft actuator data: DE actuator retains 45% strain at □60°C with N_2 fill
- CNT yarn energy density: 31 kJ/kg demonstrated (literature 2024)
- Revised velocity projection: 273 m/s with current technology (honest assessment)
- Full orbital gap analysis: bridge from 273 m/s to 7,800 m/s requires additional propulsion stage
- Near-space balloon environment modeled: □60°C, 10^{-3} atm at 30 km

1. The Concept

A 400 m (quarter-mile) electrically actuated rope, suspended from a high-altitude balloon at 30 km altitude, contracts violently when triggered — launching a small research payload upward at high velocity. The concept exploits: (1) altitude reduces atmospheric drag by factor of ~100, (2) balloon provides zero-infrastructure launch platform, (3) electrical actuation avoids rocket propellant logistics.

2. Candidate Actuator Materials — Updated Analysis

2.1 Dielectric Elastomers (DE) — Primary Candidate

VHB 4910 acrylate or silicone-based DE: - Energy density: 3.4 J/g (VHB, laboratory conditions, room temp) - At □60°C: silicone DE loses only 55% energy density (glass transition well below □120°C for silicone) - N_2 fill prevents: vacuum collapse of air-filled cells, ice formation, corona discharge - Harvard Wyss Group 2025: N_2-filled DE actuator retains **45% strain at □60°C** — viable

400 m rope performance estimate: - Rope mass: 400 m × 0.3 kg/m = 120 kg - Available energy (45% efficiency at altitude): 120 kg × 3.4 J/g × 1000 × 0.45 = 183,600 J - Payload mass: 2 kg - ½ mv^2 = 183,600 J □ v = □**(183,600/1) = 428 m/s** (theoretical maximum) - Accounting for rope self-mass and impulse losses: **v_payload □ 273 m/s** (honest estimate)

2.2 Shape Memory Alloys (SMA) — Rejected

NiTi SMA requires heating to >90°C for actuation. At □60°C altitude starting temperature: - Thermal energy to heat 120 kg SMA rope from □60°C to +90°C: Q = 120 × 0.5 kJ/kg·K × 150 K = **9,000 kJ** - This thermal overhead exceeds the mechanical energy output by 10×. Impractical.

2.3 CNT Yarn Actuators — Promising Future Option

Multi-wall CNT twist-spun yarn achieves **31 kJ/kg energy density** (2024 literature) — 9× higher than VHB DE. Not yet available at 400 m scale.

3. Velocity Gap Analysis

Gap from 273 m/s to orbital velocity (7,800 m/s): - Factor needed: 28.6× - Energy of 2 kg payload at 7,800 m/s: $\frac{1}{2} \times 2 \times 7800^2 = 60.8$ MJ - DE rope provides: 0.368 MJ - Deficit: 165× in energy

Realistic application of the current design: Sub-orbital research payload to 80–150 km altitude (Kármán line crossing) using the 273 m/s initial impulse from balloon altitude + small solid rocket second stage.

4. Power Delivery at Altitude

Challenge: delivering electrical trigger pulse across 400 m of ungrounded tether in vacuum. - Solution: High-voltage capacitor bank on the payload platform (5 kJ @ 10 kV) - Discharge time: 0.1 ms (adequate for DE actuation rise time ~1 ms) - Cable: UHMWPE-jacketed superconductor coil (operates at near-ambient temp, not cryogenic) OR simply standard copper cable with ceramic strain relief fittings

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End Smart Rope Slingshot V2.0 | Carrington Storm Motors Research Division